Valuation Mock Final

Updated November 1, 2026

Eight original syllabus-based questions, 100 points, suggested 120 minutes. Questions and writing space precede separate worked solutions, beginner warm-ups and marking guides.

How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

Valuation mock final

8 questions | 100 points | 120 minutes suggested practice time.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Derive an increasing cash-flow value (12 points)
2. Bootstrap and check interest-rate sensitivity (12 points)
3. Price an option using a cash-flow risk adjustment (12 points)
4. Audit a multivariate simulation (12 points)
5. Read a probability clue from option prices (12 points)
6. Compare committing now with waiting (14 points)
7. Connect financing assumptions to capital budgeting (14 points)
8. Trace losses and choose a risk response (12 points)

Question 1: Derive an increasing cash-flow value

12 points | Revised syllabus week 1; historical midterm derivation style.

Cash flows at years 1, 2, 3, ... are $10, $20, $30, ... forever. Derive V using a shift and GVE, then evaluate it at r = 10% annually. Explain why using the same rate for uncertain future cash flows needs an additional assumption.

Show your setup, units, calculation and interpretation.

Question 2: Bootstrap and check interest-rate sensitivity

12 points | Revised syllabus weeks 2-3: bonds, zero curve and interest-rate risk.

The one-year annual spot rate is 4%. A two-year annual-coupon bond with face $100 and coupon 5% costs $100. Derive D2 and the two-year annually compounded spot rate. Separately a bond portfolio is worth $20,000 with modified duration 4.2. Estimate its value change if its yield rises by 0.50 percentage points. State what you are ignoring.

Show your setup, units, calculation and interpretation.

Question 3: Price an option using a cash-flow risk adjustment

12 points | Revised syllabus weeks 6 and 9: Lintner CAPM and option valuation.

There are two equally likely physical states. Market cash flow XM is 80 or 120; asset cash flow X is 70 or 130 in the same states. A call pays Y = max(X - 100, 0). Use V(Y) = [E(Y) - A Cov(Y,XM)]/(1 + rf), with A = 0.01 and rf = 5%. Find the call’s expected cash flow, covariance, risk-adjusted value and value if the covariance charge were wrongly omitted. Explain the difference.

Show your setup, units, calculation and interpretation.

Question 4: Audit a multivariate simulation

12 points | Revised syllabus weeks 4-5; historical midterm computational style.

Find a lower-triangular C for covariance matrix [[9,6],[6,20]]. State the transformation from independent standard-normal rows U to correlated rows W and its inverse. A separate simulation gives payoff standard deviation s = 24 from n = 144 independent draws. With maxerr = 2SE, find its maxerr and the number of draws needed for maxerr no greater than 1, keeping s unchanged.

Show your setup, units, calculation and interpretation.

Question 5: Read a probability clue from option prices

12 points | Revised syllabus week 10: implied probability distributions.

Same-expiry European calls on a non-dividend stock have prices C(90) = 16, C(100) = 9, C(110) = 4. The expiry discount factor is D = 0.95. Find the cost of the 90/100/110 butterfly, its maximum terminal payoff, and a centered finite-difference estimate of the risk-neutral density near strike 100 using $fQ(K) \approx [C(K-h) - 2C(K) + C(K+h)]/(D\,h^2)$, h = 10. Interpret fQ × h carefully. What fails if C(100) were 11 instead?

Show your setup, units, calculation and interpretation.

Question 6: Compare committing now with waiting

14 points | Revised syllabus week 11: contracts and real options.

Next year a project asset will be worth 160 or 80. Treat these as market-consistent asset values with risk-neutral probabilities 0.5 each. The annual risk-free growth factor is 1.05. You can either pay 100 now to own the asset next year, or hold a free right to pay 100 next year after seeing its value. Find immediate-investment NPV, the value of waiting and the incremental flexibility value. Explain why using this method for a nontraded business needs care.

Show your setup, units, calculation and interpretation.

Question 7: Connect financing assumptions to capital budgeting

14 points | Revised syllabus weeks 11-12: capital budgeting and capital structure.

A company has market equity value 600 and debt 400. Its cost of equity is 12%, pre-tax cost of debt is 6%, and corporate tax rate is 25%. Under a stable target financing mix and comparable project risk, calculate after-tax WACC. A project costs 100 now and pays one unlevered after-tax cash flow of 109 next year. Find NPV and IRR, and explain the decision at this rate. Why can you not keep the same component costs while increasing debt indefinitely to make value rise?

Show your setup, units, calculation and interpretation.

Question 8: Trace losses and choose a risk response

12 points | Revised syllabus: corporate bonds/CDOs and later corporate risk management.

A $100 debt pool has equity 5, mezzanine 10 and senior 85. Allocate a realized $12 pool loss. Separately a company’s annual loss distribution is 0 with probability 90%, 10 with probability 8%, and 40 with probability 2% (all in $m). Find expected loss, 95% VaR as the smallest loss with cumulative probability at least 95%, and ES as the average of the worst 5%, splitting the mass at the VaR threshold as needed. Give one reason management should examine stress scenarios as well.

Show your setup, units, calculation and interpretation.

Solution 1: Derive an increasing cash-flow value

First, the words

GVE links today's value to the next cash flow and remaining value. A perpetuity continues indefinitely; the discount rate here is positive and per year.

A smaller problem first

A flat $10 perpetuity beginning next year is worth 10/r. Use a shifted stream to handle payments that grow by a fixed dollar amount.

Hint

After the first year, the remaining stream is the original stream plus $10 each year.

Work through it

1. Let Vnext value the stream 20,30,40,... one year later. Relative to 10,20,30,... it adds a $10 perpetuity, so Vnext = V + 10/r.

2. GVE: (1 + r)V = 10 + Vnext = 10 + V + 10/r.

3. Therefore rV = 10 + 10/r and V = 10(1 + r)/r^2.

4. At r = 0.10, V = 10 × 1.1/0.01 = $1,100. The first payment is at year 1 and r > 0 ensures convergence.

5. For uncertain cash flows, their risk and timing must be priced consistently. A risk-free rate applied to raw expected risky cash flows generally omits a risk adjustment.

Mark your reasoning

1. 4: shifted stream

2. 3: GVE and formula

3. 3: value, timing and domain

4. 2: risk limitation

Shimko Valuation Principles Chapters 1-2 and Lecture 1.

My mark: ____ / 12. My next correction: _____

Solution 2: Bootstrap and check interest-rate sensitivity

First, the words

Bootstrapping finds each discount factor from traded cash flows. Duration is a first-order approximation, not an exact prediction for large yield moves.

A smaller problem first

A one-year zero paying $100 at 4% annually costs 100/1.04. That gives D1 = 1/1.04.

Hint

Price = 5D1 + 105D2. For duration, convert 0.50 percentage points to 0.005.

Work through it

1. D1 = 1/1.04 = 0.96153846.

2. 100 = 5D1 + 105D2, so D2 = (100 - 5/1.04)/105 = 0.90659341.

3. Two-year spot rate z2 = D2^(-1/2) - 1 = 5.02525% annually.

4. Approximate change = -modified duration × value × yield change = -4.2 × 20,000 × 0.005 = -$420; estimated new value $19,580.

5. The approximation ignores convexity and changes in cash flows; a single yield shift also does not capture arbitrary curve twists or spread changes.

Mark your reasoning

1. 4: bootstrapping setup

2. 3: D2 and spot rate

3. 3: duration calculation

4. 2: limitations

Valuation bond/interest-rate lecture; zero-coupon term-structure mini-project topics.

My mark: ____ / 12. My next correction: _____

Solution 3: Price an option using a cash-flow risk adjustment

First, the words

An option’s cash flow is nonlinear. Compute its own covariance with the market before applying the supplied cash-flow pricing model.

A smaller problem first

A call paying max(X - 100, 0) on asset cash flows 70 and 130 pays 0 and 30, not -30 and 30.

Hint

Covariance is the average of the products of deviations from each variable’s mean.

Work through it

1. Y is 0 in the low state and 30 in the high state. E(Y) = 15; E(XM) = 100.

2. Cov(Y,XM) = 0.5(-15)(-20) + 0.5(15)(20) = 300.

3. Risk charge = 0.01 × 300 = 3. Call value = (15 - 3)/1.05 = $11.4286.

4. Ignoring risk gives 15/1.05 = $14.2857, an overstatement of $2.8571 in this stipulated model.

5. The option pays in the high-market state, so the model charges for positive market covariance. This is a given one-period model, not an assertion that physical probabilities equal risk-neutral probabilities.

Mark your reasoning

1. 2: payoffs and mean

2. 4: covariance

3. 3: adjusted price

4. 3: comparison and interpretation

Valuation Lecture 6 and revised syllabus week 9; follow your instructor’s notation and calibration.

My mark: ____ / 12. My next correction: _____

Solution 4: Audit a multivariate simulation

First, the words

Cholesky creates correlated draws. Standard error measures uncertainty in an estimated mean; it is different from the standard deviation of individual outcomes.

A smaller problem first

For $C = [[a,0],[b,c]]$, $CC^T = [[a^2,ab],[ab,b^2+c^2]]$. With observations in rows, transform independent draws using C^T.

Hint

Match a^2, ab and $b^2 + c^2$. Then use SE = s/sqrt(n).

Work through it

1. a = 3, b = 2, c = 4. Thus $C = [[3,0],[2,4]]$. Check $CC^T = [[9,6],[6,20]]$.

2. For row observations $W = UC^T$; recover $U = W(C^T)^{-1}$, after subtracting means if needed.

3. SE = 24/sqrt(144) = 2. Therefore maxerr = 4.

4. Need $2 \times 24/\sqrt{n} \leq 1$, so $n \geq 48^2 = 2{,}304$.

5. Independence and unchanged outcome spread are assumptions. More draws reduce sampling error, not model bias or the underlying payoff risk.

Mark your reasoning

1. 4: Cholesky and check

2. 3: row transformation and inverse

3. 3: error and sample size

4. 2: limitations

Valuation simulation lecture and Cholesky spreadsheet; historical maxerr convention is specified explicitly.

My mark: ____ / 12. My next correction: _____

Solution 5: Read a probability clue from option prices

First, the words

A butterfly combines calls at three equally spaced strikes. Call-price curvature contains information about risk-neutral probabilities, not directly real-world probabilities.

A smaller problem first

A long butterfly buys the low-strike call, sells two middle-strike calls and buys the high-strike call. Its payoff is never negative.

Hint

Positive curvature is required by no arbitrage. A density has units of probability per dollar.

Work through it

1. Butterfly cost = 16 - 2 × 9 + 4 = $2.

2. The payoff rises from 0 at S = 90 to $10 at S = 100 and falls to 0 at S = 110; outside that range it is zero.

3. Density estimate = 2/(0.95 × 100) = 0.0210526 per dollar.

4. Multiplying by a $10 interval gives roughly 0.2105, or 21.05%, only if a local constant-density approximation is reasonable. This coarse estimate is not an exact interval probability.

5. If the middle call were 11, butterfly cost would be -$2 while terminal payoff remains nonnegative. That violates convexity and allows an arbitrage under frictionless trading.

6. These are pricing-measure probabilities. Risk premia and the finite strike spacing prevent reading them as a precise real-world forecast.

Mark your reasoning

1. 3: cost and payoff

2. 3: density calculation

3. 3: careful interpretation

4. 3: convexity violation

Valuation implied-density and volatility-surface lectures; the formula is supplied to make the exercise self-contained.

My mark: ____ / 12. My next correction: _____

Solution 6: Compare committing now with waiting

First, the words

An option to invest lets you reject a bad outcome. A risk-neutral probability is given here; do not substitute physical forecasts without changing the model.

A smaller problem first

If next year’s project asset is worth 80 and investment costs 100, an optional investment pays 0 because you can decline.

Hint

Discount the asset value for the commit-now case; discount max(asset value - investment cost, 0) for the waiting case.

Work through it

1. Asset PV = (0.5 × 160 + 0.5 × 80)/1.05 = $114.2857.

2. Immediate-investment NPV = 114.2857 - 100 = $14.2857.

3. Waiting payoffs are $60 in the high state and $0 in the low state.

4. Waiting value = (0.5 × 60 + 0.5 × 0)/1.05 = $28.5714.

5. Incremental value relative to committing now = 28.5714 - 14.2857 = $14.2857. It includes the ability to avoid bad investment and defer the expenditure.

6. For a nontraded project, replication may be unavailable. Justify the pricing probabilities/risk adjustment, possible competition and any cost of waiting. The supplied assumptions make this toy model calculable.

Mark your reasoning

1. 3: asset PV and immediate NPV

2. 4: contingent payoffs and option value

3. 3: incremental comparison

4. 4: assumptions and nontraded limitation

Valuation real-options-in-energy lecture; Welch capital budgeting and option intuition.

My mark: ____ / 14. My next correction: _____

Solution 7: Connect financing assumptions to capital budgeting

First, the words

WACC weights financing costs by market values. The tax adjustment here assumes deductible interest and a usable corporate tax shield.

A smaller problem first

If 60% of funding is equity and 40% is debt, multiply each financing cost by its weight; do not average them equally.

Hint

WACC = wE ke + wD kd(1 - tax). One-period IRR solves cost = future cash/(1 + IRR).

Work through it

1. Equity and debt weights are 0.6 and 0.4. After-tax debt cost is 6% × 0.75 = 4.5%.

2. WACC = 0.6 × 12% + 0.4 × 4.5% = 9%.

3. NPV = -100 + 109/1.09 = 0. IRR = 109/100 - 1 = 9%.

4. The project exactly earns the assumed required return: it is financially break-even at these inputs, not a strictly positive-NPV opportunity.

5. Higher leverage changes equity/debt risk and their required returns. Distress costs, tax-shield limits and financing constraints can alter value.

6. Use cash flows and discount rates consistently. Do not subtract interest again from the unlevered cash flow when discounting at WACC.

Mark your reasoning

1. 4: weights and WACC

2. 3: NPV

3. 2: IRR

4. 2: break-even decision

5. 3: leverage and consistency

Welch Corporate Finance: capital budgeting, financing and capital structure. This is a syllabus extension, not a historical midterm replica.

My mark: ____ / 14. My next correction: _____

Solution 8: Trace losses and choose a risk response

First, the words

A tranche absorbs losses according to seniority. A risk measure summarizes a distribution; it does not remove exposure.

A smaller problem first

If the equity tranche absorbs the first $5 of loss, a $3 pool loss never reaches mezzanine or senior investors.

Hint

The worst 5% includes all of the 2% at loss 40 and 3 percentage points from the loss-10 outcomes.

Work through it

1. Equity loses 5, mezzanine loses 7, senior loses 0. Remaining tranche values under this simple loss-only allocation are 0, 3 and 85.

2. Expected loss = 0.90 × 0 + 0.08 × 10 + 0.02 × 40 = $1.6m.

3. Cumulative probability is 90% at 0 and 98% at 10, so 95% VaR = $10m.

4. Worst-5% average = (0.02 × 40 + 0.03 × 10)/0.05 = $22m.

5. Stress testing can examine losses beyond an estimated distribution, correlation breakdowns, illiquidity and funding pressures. It complements a quantified limit rather than proving the model correct.

Mark your reasoning

1. 3: waterfall

2. 2: expected loss

3. 2: VaR

4. 3: ES

5. 2: stress rationale

Valuation corporate bonds/CDOs lecture; later corporate-risk-management syllabus topic. Risk course assessment remains report/presentation based.

My mark: ____ / 12. My next correction: _____